

MCS-218 CHEATSHEET

COMPUTER NETWORKS - GUESS QUESTIONS

Q1.1 ★★☆☆ CRC Calculation (Cyclic Redundancy Check)

Question: Find CRC for the frame 1101011011 with generator polynomial $x^4 + x + 1$.

Understanding CRC:

CRC is an error-detecting code commonly used in digital networks and storage devices to detect accidental changes to raw data. It works by treating the data as a large binary number, dividing it by a predetermined polynomial, and using the remainder as the CRC value.

Step-by-Step Solution:

Step 1: Convert Generator Polynomial to Binary

- The polynomial $x^4 + x + 1$ represents powers of x that have coefficient 1
- x^4 contributes to bit position 4 (leftmost)
- x^1 contributes to bit position 1
- x^0 (constant 1) contributes to bit position 0
- Therefore: $x^4 + x + 1 \rightarrow 10011$ (5 bits total, degree = 4)

Step 2: Append Zeros to Original Data

- Original data frame: 1101011011 (10 bits)
- Number of zeros to append = degree of polynomial = 4
- Augmented data: 11010110110000 (14 bits)

Step 3: Perform Binary Division using XOR Operation

The division process uses XOR (exclusive OR) instead of normal subtraction. When the leading bit is 1, we XOR with the divisor; when it's 0, we XOR with all zeros.

Dividend: 11010110110000 Divisor: 10011
Step 1: 11010110110000 10011
———— 1001 (XOR result)

Step 2: 10011 10011 ————— 0000 (XOR result)

Step 3: 00001011 1011 (bring down next bits)

Step 4: 10110 10011 ————— 0101 (XOR result)

Step 5: 1010 1001 (shift divisor) ——— 011 (continue...)

Final Remainder: 1110

Detailed Division Table:

Step	Current Dividend	Divisor Applied	XOR Result	Action
1	11010	10011	01001	Bring down next bit
2	10011	10011	00000	Bring down next bit
3	00001	00000	00001	Leading 0, shift
4	10110	10011	00101	Continue division
5	01011	00000	01011	Leading 0, shift
6	10110	10011	00101	Continue
7	Final Remainder	-	1110	CRC obtained

Step 4: Construct Transmitted Frame

- CRC (Remainder):** 1110
- Transmitted Frame:** Original Data + CRC = 11010110111110 (14 bits total)

Step 5: Verification at Receiver The receiver divides the entire received frame (data + CRC) by the same generator polynomial. If the remainder is zero, no error is detected. If non-zero, an error has occurred during transmission.

Common Generator Polynomials:

Name	Polynomial	Binary	Usage
CRC-4	$x^4 + x + 1$	10011	ITU-T
CRC-8	$x^8 + x^2 + x + 1$	100000111	ATM
CRC-16	$x^{16} + x^{15} + x^2 + 1$	1100000000000101	USB, Modbus
CRC-32	$x^{32} + x^{26} + \dots + 1$	(32 bits)	Ethernet, ZIP

Q1.2 ★★☆☆ Line Encoding Schemes

Question: Draw Manchester and Differential Manchester encoding for bit sequence 001100110011.

Understanding Line Encoding:

Line encoding is the process of converting digital data into digital signals. Different encoding schemes offer various advantages in terms of synchronization, bandwidth efficiency, and DC balance.

Encoding Rules Summary:

Encoding Scheme	Bit 0 Representation	Bit 1 Representation	Key Feature
NRZ-L	High voltage	Low voltage	Simple, no self-clocking
NRZ-I	No transition	Transition at start	Differential encoding

Encoding Scheme	Bit 0 Representation	Bit 1 Representation	Key Feature
Manchester	High-to-Low transition (↓)	Low-to-High transition (↑)	Transition at mid-bit
Differential Manchester	Transition at bit start	No transition at start	Always mid-bit transition
RZ (Return to Zero)	Positive pulse, return to zero	Negative pulse, return to zero	Three voltage levels

Detailed Encoding for 001100110011:

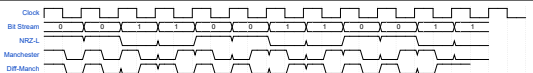
Manchester Encoding Rules:

- Bit 0: Signal goes from HIGH to LOW at the middle of the bit period
- Bit 1: Signal goes from LOW to HIGH at the middle of the bit period
- Every bit has a guaranteed transition at the midpoint (self-clocking)

Differential Manchester Encoding Rules:

- Always a transition at the MIDDLE of every bit (for clock synchronization)
- Bit 0: Additional transition at the BEGINNING of the bit period
- Bit 1: NO transition at the beginning of the bit period
- The encoding is differential (depends on previous state)

Waveform Diagram:



Step-by-Step Manchester Encoding:

Bit Position	Data Bit	Manchester Signal	Transition Direction
1	0	High→Low	Falling edge at center
2	0	High→Low	Falling edge at center
3	1	Low→High	Rising edge at center
4	1	Low→High	Rising edge at center
5	0	High→Low	Falling edge at center
6	0	High→Low	Falling edge at center
7	1	Low→High	Rising edge at center
8	1	Low→High	Rising edge at center
9	0	High→Low	Falling edge at center
10	0	High→Low	Falling edge at center
11	1	Low→High	Rising edge at center
12	1	Low→High	Rising edge at center

Comparison of Encoding Schemes:

Feature	Manchester	Differential Manchester	NRZ
Self-clocking	Yes	Yes	No
DC Component	None (balanced)	None (balanced)	Present
Bandwidth Required	2× NRZ	2× NRZ	Lower
Error Detection	Easy	Easier	Difficult
Complexity	Moderate	Higher	Simple
Used In	10BASE-T Ethernet	Token Ring	Serial comm

Advantages of Manchester Encoding:

- Self-synchronization:** Clock can be extracted from data
- No DC component:** Average voltage is zero
- Error detection:** Missing transition indicates error
- Simple implementation:** Easy to encode and decode

Q1.3 ★★☆☆ ALOHA Protocol and Throughput Analysis

Question: Define vulnerable period, derive throughput expression for pure ALOHA, and differentiate pure ALOHA with slotted ALOHA.

Introduction to ALOHA:

ALOHA is one of the earliest random access protocols developed at the University of Hawaii in 1970. It allows multiple stations to share a common broadcast channel by transmitting whenever they have data, with a mechanism to handle collisions.

Vulnerable Period Definition:

The **vulnerable period** (also called collision window) is the time interval during which the frames of other stations can potentially collide with the frame being transmitted by a given station. It represents the critical time window where any other transmission would result in a collision.

Pure ALOHA - Detailed Analysis:

Operating Principle:

- Any station can transmit at any time when it has data

- If collision occurs (detected by lack of acknowledgment), wait a random time and retransmit

- No coordination between stations

Vulnerable Period Calculation:

Consider a frame that takes time T to transmit:

- A collision occurs if any other station starts transmitting within time T BEFORE our transmission starts (their frame overlaps with beginning of ours)
- A collision also occurs if any other station starts transmitting within time T AFTER our transmission starts (our frame overlaps with beginning of theirs)
- Total Vulnerable Period = $T + T = 2T$**

Throughput Derivation for Pure ALOHA:

Assumptions:

- Infinite number of stations
- Poisson distribution for frame arrivals
- G = average number of frames generated per frame time (offered load)
- All frames have equal length requiring time T for transmission

Derivation Steps:

- For successful transmission, no other frame should begin during vulnerable period $2T$
- Using Poisson distribution, probability of k arrivals in time t :

$$P(k) = \frac{(Gt)^k \cdot e^{-Gt}}{k!}$$

- Probability of zero frames in vulnerable period $2T$:

$$P(0) = e^{-2G}$$

- Throughput S** = Offered load $\times$ Probability of success:

$$S = G \times e^{-2G}$$

Finding Maximum Throughput:

To find maximum, take derivative and set to zero:

$$\frac{dS}{dG} = e^{-2G} - 2G \cdot e^{-2G} = 0$$

$$e^{-2G}(1 - 2G) = 0$$

$$G = 0.5$$

Maximum Throughput:

$$S_{max} = 0.5 \times e^{-1} = 0.5 \times 0.368 = \boxed{0.184 = 18.4\%}$$

Slotted ALOHA - Detailed Analysis:

Operating Principle:

- Time is divided into discrete slots of duration T (frame transmission time)
- Stations can only begin transmission at the start of a time slot
- Requires synchronization among all stations

Vulnerable Period = T (only the same slot can cause collision)

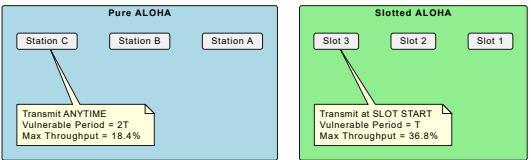
Throughput Formula:

$$S = G \times e^{-G}$$

Maximum Throughput (at $G = 1$):

$$S_{max} = 1 \times e^{-1} = \boxed{0.368 = 36.8\%}$$

Comparison Diagram:



Comprehensive Comparison Table:

Parameter	Pure ALOHA	Slotted ALOHA
Transmission Time	Any instant	Slot boundaries only
Vulnerable Period	$2T$	T
Throughput Formula	$S = G \times e^{-(2G)}$	$S = G \times e^{-(G)}$
Maximum Throughput	18.4% at $G=0.5$	36.8% at $G=1$
Synchronization	Not required	Required (clock)
Implementation	Simpler	More complex
Collision Probability	Higher	Lower
Channel Utilization	Poor	Better
Delay	Lower (immediate TX)	Higher (wait for slot)
Developed	1970	1972

Throughput vs. Offered Load Graph Description:

- Both curves start at origin and rise initially
- Pure ALOHA peaks at $G=0.5$ with $S=0.184$
- Slotted ALOHA peaks at $G=1.0$ with $S=0.368$
- Both curves decline after peak (too many collisions)
- Slotted ALOHA consistently outperforms Pure ALOHA

Q1.4 ★★☆☆ Digital Modulation and PCM

Question: Draw constellation diagram for 4-PSK and 4-QAM. Explain PCM process.

Introduction to Digital Modulation:

Digital modulation is the process of converting digital data into analog signals for transmission over a communication channel. The three fundamental properties that can be modified are amplitude, frequency, and phase.

4-PSK (Quadrature Phase Shift Keying):

Concept: In 4-PSK (also called QPSK), the carrier signal's phase is shifted to one of four distinct values to represent 2-bit combinations.

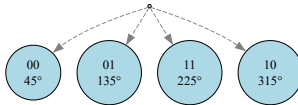
Phase Assignments:

Bit Pattern	Phase Angle	I Component	Q Component
00	45°	+1	+1
01	135°	-1	+1
10	315°	+1	-1
11	225°	-1	-1

Characteristics of 4-PSK:

- Bits per symbol:** 2 bits
- Number of phase states:** 4 (0°, 90°, 180°, 270° or 45°, 135°, 225°, 315°)
- Amplitude:** Constant (all points on circle)
- Bandwidth efficiency:** 2 bits/Hz
- Application:** Satellite communication, CDMA, DVB-S

4-PSK Constellation Diagram:



4-QAM (Quadrature Amplitude Modulation):

Concept: In 4-QAM, both amplitude and phase of the carrier are varied to create distinct signal points. Each point represents a unique 2-bit combination.

Signal Point Assignments:

Bit Pattern	I Component	Q Component	Amplitude	Phase
00	+1	+1	$\sqrt{2}$	45°
01	-1	+1	$\sqrt{2}$	135°
10	+1	-1	$\sqrt{2}$	315°
11	-1	-1	$\sqrt{2}$	225°

Note: For 4-QAM, the constellation is identical to 4-PSK. The difference becomes apparent in higher-order modulations (16-QAM, 64-QAM) where multiple amplitude levels are used.

4-QAM Constellation Diagram:

00

01

10

11

I (In-phase)

Q (Quadrature)

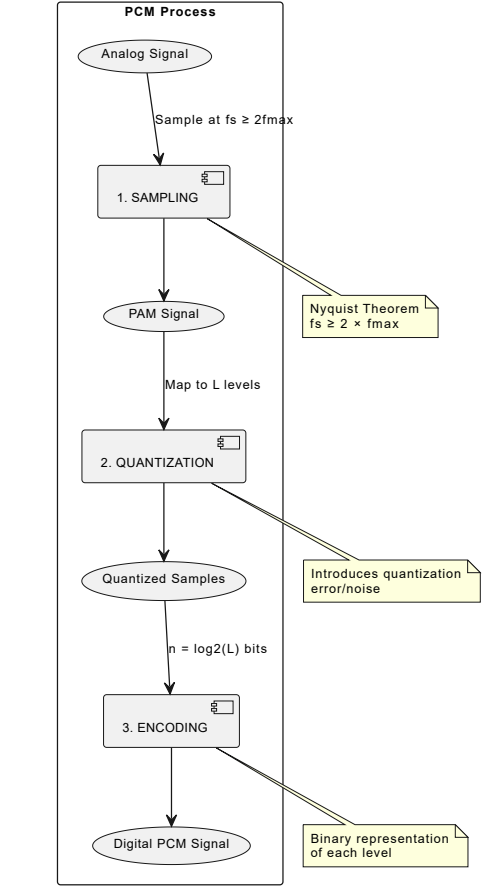
Comparison: PSK vs QAM:

Feature	PSK	QAM
Varied Parameter	Phase only	Amplitude + Phase
Constant Envelope	Yes	No
Power Efficiency	Better	Moderate
Noise Immunity	Higher	Lower
Spectral Efficiency	Moderate	Higher (for higher orders)
Complexity	Lower	Higher

PCM (Pulse Code Modulation) - Detailed Explanation:

PCM is a method used to digitally represent analog signals. It is the standard form of digital audio in computers, compact discs, and digital telephony.

Three Main Steps of PCM:



- Step 1: Sampling**
- The continuous analog signal is sampled at regular intervals
 - Nyquist Theorem:** Sampling frequency must be at least twice the maximum signal frequency
 - $fs \geq 2 \times fmax$ (to avoid aliasing)
 - Example: For telephone quality audio ($fmax = 4\text{ kHz}$), $fs = 8\text{ kHz}$
- Step 2: Quantization**
- Each sample is approximated to the nearest discrete level
 - Number of quantization levels = $L = 2^n$ (where n = number of bits)

- Quantization Error:** Difference between actual and quantized value
- More levels = better quality but more bits required

- Step 3: Encoding**
- Each quantized level is converted to a binary code
 - Number of bits per sample = $n = \log_2(L)$
 - Example: 256 levels require 8 bits per sample

PCM Formulas and Calculations:

Parameter	Formula	Example (Telephone)
Min. Sampling Rate	$fs \geq 2 \times fmax$	$2 \times 4000 = 8000\text{ Hz}$
Quantization Levels	$L = 2^n$	$2^8 = 256\text{ levels}$
Bits per Sample	$n = \log_2(L)$	$\log_2(256) = 8\text{ bits}$
Bit Rate	$R = fs \times n$	$8000 \times 8 = 64\text{ kbps}$
Signal-to-Quantization Noise	$SQNR = 6.02n + 1.76\text{ dB}$	$6.02(8) + 1.76 = 49.92\text{ dB}$

PCM Applications:

- Digital telephone systems (64 kbps per channel)
- CD audio (44.1 kHz, 16-bit, stereo = 1.4 Mbps)
- Digital audio broadcasting
- Voice over IP (VoIP)

Q1.5 ★★★★★ OSI and TCP/IP Reference Models

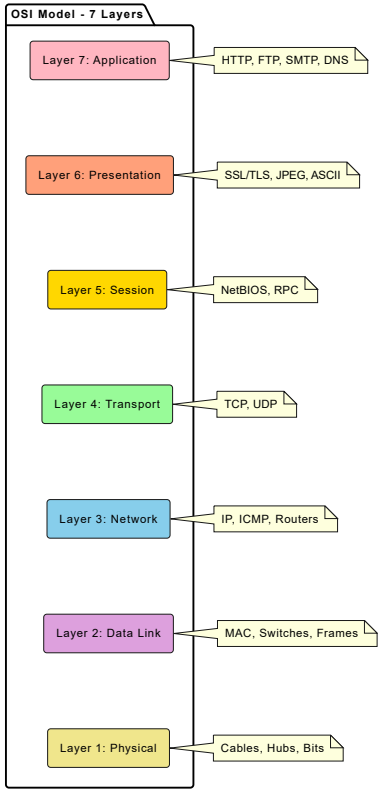
Question: List and explain functionalities of OSI reference model layers. Compare OSI and TCP/IP.

Introduction to Network Models:

Network models provide a structured framework for understanding and implementing network communications. They divide the complex task of network communication into manageable layers, each with specific responsibilities.

OSI Reference Model (7 Layers):

The Open Systems Interconnection (OSI) model was developed by the International Organization for Standardization (ISO) in 1984. It serves as a universal reference model for network communication.



Detailed Layer Functions:

Layer 7 - Application Layer:

- Provides interface between user applications and network services
- Handles network transparency, resource allocation
- Protocols:** HTTP (web), FTP (file transfer), SMTP (email), DNS (naming), Telnet, SSH
- PDU:** Data/Message
- Function:** Network virtual terminal, file transfer, mail services, directory services

Layer 6 - Presentation Layer:

- Translates data between application and network formats
- Handles data encryption, decryption, and compression
- Functions:** Character code translation (ASCII↔EBCDIC), data encryption (SSL/TLS), data compression, format conversion (JPEG, MPEG, GIF)
- PDU:** Data
- Also called "Syntax Layer"

Layer 5 - Session Layer:

- Establishes, maintains, and terminates communication sessions
- Provides dialog control and synchronization
- Functions:** Session establishment, maintenance, termination; Dialog control (half-duplex/full-duplex); Synchronization with checkpoints
- Protocols:** NetBIOS, RPC, SQL
- PDU:** Data

Layer 4 - Transport Layer:

- Provides end-to-end communication between processes
- Ensures complete data transfer with error recovery
- Functions:** Segmentation and reassembly; Connection-oriented (TCP) or connection-less (UDP); Flow control; Error control; Multiplexing
- Protocols:** TCP (reliable), UDP (fast), SCTP
- PDU:** Segment (TCP) / Datagram (UDP)
- Devices:** Gateway

Layer 3 - Network Layer:

- Handles logical addressing and routing between different networks
- Determines best path for data transmission
- Functions:** Logical addressing (IP); Routing (path determination); Packet forwarding; Fragmentation and reassembly; Congestion control
- Protocols:** IP, ICMP, OSPF, BGP, RIP
- PDU:** Packet
- Devices:** Router, Layer 3 Switch

Layer 2 - Data Link Layer:

- Provides node-to-node data transfer and error detection
- Divided into LLC (Logical Link Control) and MAC (Media Access Control) sublayers
- Functions:** Framing; Physical addressing (MAC); Error detection (CRC); Flow control; Media access control
- Protocols:** Ethernet, Wi-Fi (802.11), PPP, HDLC
- PDU:** Frame
- Devices:** Switch, Bridge, NIC

Layer 1 - Physical Layer:

- Handles physical transmission of raw bits over the medium
- Defines electrical, mechanical, and procedural specifications
- Functions:** Bit synchronization; Bit rate control; Physical topology; Transmission mode (simplex, duplex); Encoding schemes
- Standards:** RS-232, V.35, Ethernet physical, DSL
- PDU:** Bits
- Devices:** Hub, Repeater, Cables, Modem

TCP/IP Model (4 Layers):

Developed by DARPA for the ARPANET, TCP/IP is the practical model used on the Internet.

Layer	TCP/IP Name	Equivalent OSI Layers	Protocols
4	Application	Application + Presentation + Session	HTTP, FTP, DNS, SMTP
3	Transport	Transport	TCP, UDP
2	Internet	Network	IP, ICMP, ARP
1	Network Access	Data Link + Physical	Ethernet, Wi-Fi

Comprehensive OSI vs TCP/IP Comparison:

Aspect	OSI Model	TCP/IP Model
Number of Layers	7 layers	4 layers
Developed By	ISO (International Organization for Standardization)	DoD (Department of Defense) / DARPA
Year	1984	1970s (standardized 1983)
Approach	Protocol independent, generic	Protocol specific, practical

Aspect	OSI Model	TCP/IP Model
Session & Presentation	Separate dedicated layers	Combined into Application layer
Network Access	Data Link + Physical separate	Combined into single layer
Reliability	Transport layer	Transport layer
Usage	Reference/educational model	Real-world implementation
Protocol Development	Model first, then protocols	Protocols first, then model
Header Size	Generally more overhead	Optimized headers
Flexibility	More flexible (layer independence)	Less flexible
Implementation	Complex, not widely implemented	Simple, universally implemented
Connectionless/Connection	Supports both at Network layer	Only connectionless at Internet layer

Memory Aid (Mnemonic):

OSI Layers (Top to Bottom): "All People Seem To Need Data Processing"

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical

Q1.6 ★★★★★ Go-Back-N ARQ and Flow Control Mechanisms

Question: Explain pipelining in Go-Back-N ARQ. What is piggybacking?

Introduction to ARQ Protocols:

Automatic Repeat reQuest (ARQ) protocols are error-control methods for data transmission that use acknowledgments and timeouts to achieve reliable data transfer. They ensure that data is correctly received by the destination.

Types of ARQ Protocols:

- Stop-and-Wait ARQ** - Send one frame, wait for ACK
- Go-Back-N ARQ** - Send multiple frames, go back to error point
- Selective Repeat ARQ** - Send multiple frames, retransmit only errored frames

Go-Back-N ARQ - Detailed Explanation:

Concept: The sender can transmit multiple frames (up to N) without waiting for individual acknowledgments. If an error is detected or timeout occurs, the sender "goes back" and retransmits all frames starting from the errored frame.

Key Parameters:

- Sender Window Size = N** (number of frames that can be sent without ACK)
- Receiver Window Size = 1** (can only accept frames in order)
- Sequence Numbers Required:** At least $N + 1$

Pipelining in Go-Back-N:

Definition: Pipelining is the technique of sending multiple frames before receiving acknowledgment for the first frame. This keeps the communication channel busy and improves efficiency.

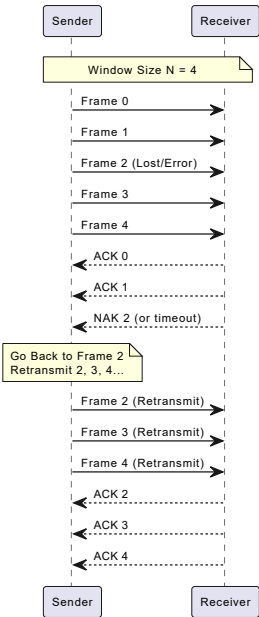
Without Pipelining (Stop-and-Wait):

- Send Frame 0 → Wait for ACK → Send Frame 1 → Wait for ACK...
- Channel remains idle during round-trip time
- Very inefficient for high bandwidth-delay product links

With Pipelining (Go-Back-N):

- Send Frame 0, 1, 2, 3... up to N frames continuously
- Receive ACKs while sending new frames
- Channel utilization significantly improved

Go-Back-N Operation Diagram:



Efficiency Analysis:

Efficiency Formula:

$$\eta = \frac{N}{1 + 2a}$$

- Where:
- N = Window size (number of frames)
 - a = Propagation delay / Transmission delay = T_p/T_t
- Conditions:
- If $N \geq (1 + 2a)$: Efficiency approaches 100%
 - If $N < (1 + 2a)$: Efficiency = $N/(1 + 2a)$

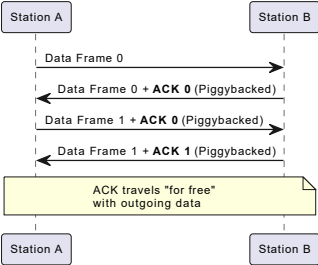
Example Calculation:

- Bandwidth = 1 Mbps, Frame size = 1000 bits, RTT = 50 ms
- $T_t = 1000/(1 \times 10^6) = 1$ ms
- $a = RTT/(2 \times T_t) = 50/(2 \times 1) = 25$
- For N = 4: $\eta = 4/(1 + 50) \approx 7.8\%$
- For N = 51: $\eta = 51/(1 + 50) = 100\%$

Piggybacking - Detailed Explanation:

Definition: Piggybacking is a technique used in bidirectional (full-duplex) communication where acknowledgment information is attached to ("piggybacks on") an outgoing data frame rather than sending a separate ACK frame.

- How Piggybacking Works:**
1. Station A sends data frame to Station B
 2. Station B has data to send to Station A
 3. Instead of sending separate ACK for A's frame, B includes the ACK in its own data frame header
 4. This reduces the total number of frames transmitted



- Benefits of Piggybacking:**
1. **Reduced Overhead:** Fewer control frames needed
 2. **Better Bandwidth Utilization:** ACK doesn't use separate frame
 3. **Improved Efficiency:** More data can be transmitted in same time
- Delayed Acknowledgment:**
- If no outgoing data available, station waits briefly for data
 - If timeout occurs, sends separate ACK
 - Typical delay: 200-500 ms

Comparison of ARQ Protocols:			
Feature	Stop-and-Wait	Go-Back-N	Selective Repeat
Sender Window	1	N	N
Receiver Window	1	1	N
Retransmission	Single frame	N frames from error	Only error frame
Efficiency	$1/(1+2a)$	$N/(1+2a)$	$N/(1+2a)$
Receiver Buffer	Minimal	Minimal	Larger (stores out-of-order)
Complexity	Simple	Moderate	Complex
Sequence Numbers	2 minimum	N + 1 minimum	2N minimum
NAK Required	Optional	Optional	Yes
Best For	Low BDP links	Moderate error rates	High error rates

Q1.7 ★★★★★ Traffic Shaping Algorithms

Question: Differentiate between Leaky Bucket and Token Bucket shaper. Explain traffic policing.

Introduction to Traffic Shaping:

Traffic shaping (also called packet shaping) is a computer network traffic management technique that delays some or all datagrams to bring them into compliance with a desired traffic profile. It is used to optimize or guarantee performance, improve latency, or increase usable bandwidth.

- Why Traffic Shaping is Needed:**
1. **Prevent Network Congestion:** Control traffic flow to avoid overwhelming network resources
 2. **QoS Guarantee:** Ensure bandwidth allocation for critical applications
 3. **Fair Bandwidth Distribution:** Prevent single users from monopolizing bandwidth
 4. **Contract Enforcement:** Ensure users comply with agreed bandwidth limits

Leaky Bucket Algorithm:

Concept: The leaky bucket algorithm works like a bucket with a small hole at the bottom. Water (packets) can be poured in at any rate, but it leaks out at a constant rate. If the bucket overflows, excess water (packets) is discarded.

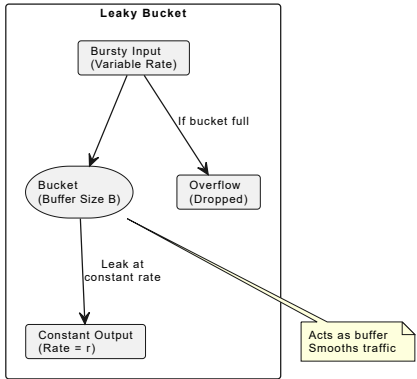
- How It Works:**
1. Packets arrive at variable rate and enter the bucket (buffer)
 2. Packets are transmitted (leak out) at a fixed, constant rate
 3. If bucket is full, incoming packets are dropped
 4. Output rate remains constant regardless of input burstiness

- Algorithm Steps:**
1. Initialize bucket with capacity B
 2. For each incoming packet:
 - If bucket has space → Add packet to bucket
 - Else → Drop packet (overflow)
 3. At regular intervals (constant rate r):
 - Remove one packet from bucket
 - Transmit the packet

Leaky Bucket Characteristics:	
Parameter	Description
Input Rate	Variable (bursty traffic allowed)
Output Rate	Constant (fixed rate r)
Bucket Size	Maximum burst that can be absorbed

Parameter	Description
Overflow Policy	Drop excess packets
Traffic Pattern	Completely smoothed output

Leaky Bucket Diagram:



Token Bucket Algorithm:

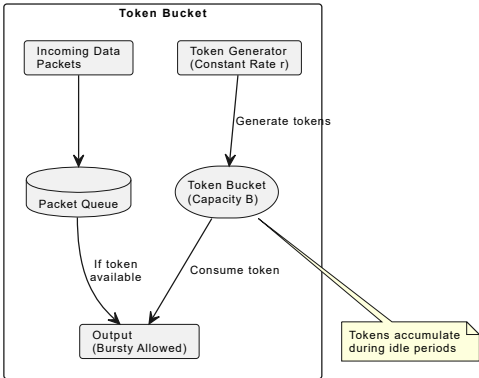
Concept: The token bucket algorithm uses tokens that are generated at a constant rate. Each packet requires a token to be transmitted. Tokens can accumulate in the bucket (up to maximum capacity), allowing controlled bursts of traffic.

- How It Works:**
1. Tokens are added to bucket at constant rate r tokens/second
 2. Bucket can hold maximum B tokens
 3. To transmit a packet, a token must be removed from bucket
 4. If no tokens available, packet must wait (shaping) or be dropped (policing)
 5. If tokens accumulate, burst transmission is allowed

- Algorithm Steps:**
1. Token generator adds tokens at rate r
 2. Maximum tokens in bucket = B
 3. For each outgoing packet:
 - If token available → Remove token, send packet
 - Else → Wait for token (shaping) OR drop packet (policing)
 4. Burst size limited by accumulated tokens

Token Bucket Characteristics:	
Token Rate	r tokens per second (constant)
Bucket Capacity	B tokens (max burst size)
Output Rate	Variable (up to B + rt in time t)
Burst Handling	Allows controlled bursts
Idle Behavior	Tokens accumulate for later use

Token Bucket Diagram:



Comprehensive Comparison:		
Feature	Leaky Bucket	Token Bucket
Output Rate	Strictly constant	Variable (allows bursts)
Burst Handling	Completely smoothed	Controlled bursts allowed
Idle Period	Capacity wasted	Tokens saved for later
Maximum Burst	None (constant output)	B tokens worth
Flexibility	Less flexible	More flexible
Token Concept	No tokens used	Uses token mechanism
Implementation	Simpler	More complex
When Full	Drop incoming packets	Wait or drop data packets
Average Rate	= leak rate	= token generation rate
Peak Rate	= leak rate	Can exceed average temporarily
Use Case	Strict rate limiting	QoS with burst tolerance
Analogy	Bucket with hole	Toll booth with tokens

Mathematical Analysis:

- Leaky Bucket:**
- Maximum output = $r \times t$ (constant rate over time t)
 - No burst capability
- Token Bucket:**
- Maximum burst size = B (bucket capacity)
 - Maximum data in time $t = B + r \times t$
 - Long-term average rate = r

Traffic Policing vs Traffic Shaping:		
Aspect	Traffic Policing	Traffic Shaping
Action on Excess	Drops packets immediately	Delays packets in queue
Buffering	No buffering	Uses buffer/queue
Latency	No added latency	May increase latency
Packet Loss	Higher (drops excess)	Lower (delays instead)
Implementation	Simpler	More complex
Use Location	Network edge (ingress)	Network edge (egress)
Resource Usage	Less memory	More memory for buffers
Effect on TCP	Can trigger retransmissions	Smoother TCP behavior
Example	ISP rate limiting	Enterprise QoS

Practical Applications:

- Leaky Bucket Used In:**
- ATM networks (constant bit rate traffic)
 - Traffic policing at network ingress
 - Rate limiting in firewalls
- Token Bucket Used In:**
- TCP/IP traffic shaping
 - Quality of Service (QoS) implementations
 - Bandwidth allocation in routers
 - API rate limiting in web services

Q1.8 ★★ Network Topologies

Question: Describe star topology. Compare star and tree topology.

Introduction to Network Topology:

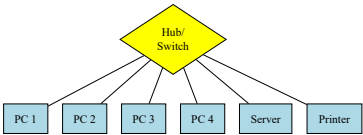
Network topology refers to the arrangement of different elements (links, nodes) in a computer network. It can be viewed as the physical layout (how devices are actually connected) or logical layout (how data flows through the network).

Star Topology - Detailed Description:

Definition: In star topology, all network devices (nodes) are connected to a central device called a hub, switch, or concentrator. All communication passes through this central device.

- Structure:**
- Central hub/switch at the center
 - All nodes connect directly to the central device
 - No direct connections between nodes
 - Data from one node passes through hub to reach another

Star Topology Diagram:



Advantages of Star Topology:

- 1. **Easy Installation:** Simple to set up and modify
- 2. **Fault Isolation:** Failure of one node doesn't affect others
- 3. **Easy Troubleshooting:** Problems can be easily isolated to specific cable or node
- 4. **Centralized Management:** Easy to monitor and manage from central point
- 5. **Scalability:** Easy to add or remove nodes without disruption
- 6. **Performance:** No collision issues with switched hub

Disadvantages of Star Topology:

- 1. **Single Point of Failure:** Hub failure brings down entire network
- 2. **Cable Requirement:** More cable needed than bus topology
- 3. **Hub Cost:** Central device adds to network cost
- 4. **Limited Distance:** Cable length limitations from hub to nodes
- 5. **Hub Capacity:** Number of nodes limited by hub port count

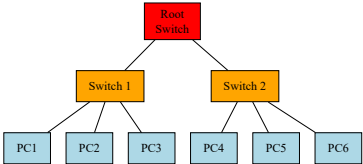
Tree Topology - Description:

Definition: Tree topology (also called hierarchical topology) combines characteristics of star and bus topologies. It has a root node and all other nodes are connected in a hierarchy.

Structure:

- Hierarchical arrangement of nodes
- Root node at top level
- Branch nodes connect to root
- Leaf nodes at bottom level
- Similar to organizational hierarchy

Tree Topology Diagram:



Advantages of Tree Topology:

- 1. **Scalability:** Easy to expand network by adding branches
- 2. **Manageability:** Hierarchical structure aids management
- 3. **Fault Isolation:** Can isolate entire branch if needed
- 4. **Point-to-Point Wiring:** Each segment is point-to-point
- 5. **Departmentalization:** Natural fit for organizational structure

Disadvantages of Tree Topology:

- 1. **Root Dependency:** Root node failure affects entire network
- 2. **Complex Configuration:** More complex than star
- 3. **Cable Usage:** High amount of cabling required
- 4. **Maintenance:** Difficult to configure and maintain
- 5. **Cost:** Higher cost due to multiple switches/hubs

Comprehensive Comparison: Star vs Tree Topology:

Feature	Star Topology	Tree Topology
Structure	Flat (single hub)	Hierarchical (multiple levels)
Central Device	Single hub/switch	Root node + multiple hubs
Scalability	Limited by hub ports	High (add branches)
Complexity	Simple	More complex
Cabling	Moderate	Extensive
Cost	Lower	Higher
Failure Impact	Hub failure = total failure	Root failure = total, branch failure = partial
Management	Centralized	Distributed hierarchical
Best For	Small to medium LANs	Large enterprise networks
Data Flow	Through single hub	Through hierarchy
Examples	Office LAN, Home network	Campus network, WAN
Expansion	Add nodes to hub	Add branches or levels

Other Common Topologies (Brief Overview):

Topology	Description	Pros	Cons
Bus	Single backbone cable	Cheap, easy	Single point failure
Ring	Circular connection	Equal access	One break stops all
Mesh	Every node connected	Redundant, reliable	Expensive, complex
Hybrid	Combination of types	Flexible	Complex to design

Q1.9 ★★ Transmission and Propagation Delay

Question: Describe transmission and propagation delays. Calculate bit rate.

Types of Network Delays:

When data travels from source to destination, it experiences four types of delays:

- 1. **Transmission Delay (Tt)**
- 2. **Propagation Delay (Tp)**
- 3. **Processing Delay**
- 4. **Queuing Delay**

Transmission Delay - Detailed Explanation:

Definition: Transmission delay is the time required to push (transmit) all bits of a packet onto the transmission link. It depends on packet size and link bandwidth.

Formula:

$$T_t = \frac{L}{B}$$

Where:

- **L** = Packet length (in bits)
- **B** = Bandwidth of the link (in bits per second)
- **Tt** = Transmission delay (in seconds)

Characteristics:

- Independent of distance
- Depends on packet size and link speed
- Faster links have lower transmission delay
- Also called "store-and-forward delay"

Example:

- Packet size L = 1000 bits
- Bandwidth B = 1 Mbps = 10⁶ bps
- Tt = 1000 / 10⁶ = 0.001 seconds = **1 millisecond**

Propagation Delay - Detailed Explanation:

Definition: Propagation delay is the time required for a signal (single bit) to travel from the sender to the receiver through the transmission medium. It depends on distance and signal speed.

Formula:

$$T_p = \frac{D}{S}$$

Where:

- **D** = Distance between sender and receiver (in meters)
- **S** = Propagation speed of signal in medium (in m/s)
- **Tp** = Propagation delay (in seconds)

Signal Speeds in Different Media:

Medium	Propagation Speed	Approximate Value
Vacuum/Air	Speed of light	3 × 10 ⁸ m/s
Fiber Optic	0.67 × c	2 × 10 ⁸ m/s
Copper Cable	0.67 × c	2 × 10 ⁸ m/s
Coaxial Cable	0.65 × c	1.95 × 10 ⁸ m/s

Example:

- Distance D = 1000 km = 10⁶ meters
- Speed S = 2 × 10⁸ m/s
- Tp = 10⁶ / (2 × 10⁸) = 0.005 seconds = **5 milliseconds**

Comparison of Delays:

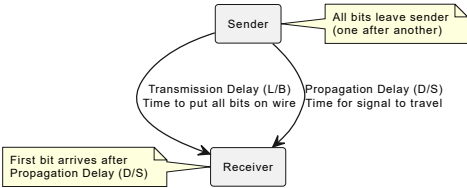
Feature	Transmission Delay	Propagation Delay
Definition	Time to push all bits onto link	Time for bit to travel to destination
Formula	L/B	D/S
Depends On	Packet size, bandwidth	Distance, medium
Independent Of	Distance	Packet size, bandwidth

Feature	Transmission Delay	Propagation Delay
Analogy	Time to load truck	Time for truck to travel
Reduction	Increase bandwidth	Use faster medium

Total End-to-End Delay:

$$T_{total} = T_{transmission} + T_{propagation} + T_{processing} + T_{queuing}$$

Delay Diagram:



Bit Rate Calculation Example:

Given Problem: An analog signal carries 4 bits per signal element. The system transmits 1000 signal elements per second. Calculate the bit rate.

Solution:

Known Values:

- Bits per signal element (baud) = 4 bits
- Signal rate (baud rate) = 1000 signals/second

Formula:

$$\text{Bit Rate} = \text{Bits per signal} \times \text{Signal rate (baud rate)}$$

Calculation:

$$\text{Bit Rate} = 4 \times 1000 = \mathbf{4000 \text{ bps (bits per second)}}$$

Related Formulas and Concepts:

Nyquist Bit Rate (Noiseless Channel):

$$C = 2 \times B \times \log_2(L)$$

Where:

- **C** = Maximum bit rate (bps)
- **B** = Bandwidth (Hz)
- **L** = Number of signal levels

Shannon Capacity (Noisy Channel):

$$C = B \times \log_2(1 + SNR)$$

Where:

- **SNR** = Signal-to-Noise Ratio

Bandwidth-Delay Product:

$$BDP = \text{Bandwidth} \times RTT$$

This represents the maximum amount of data that can be "in flight" (transmitted but not yet acknowledged) in the network.

Efficiency and Utilization:

Link Utilization:

$$U = \frac{T_t}{T_t + 2T_p} = \frac{1}{1 + 2a}$$

Where a = Tp/Tt (propagation-to-transmission delay ratio)

Q1.10 ★★ Cryptography Fundamentals

Question: Define cryptography, encryption, decryption. Why is it important for network security?

Introduction to Cryptography:

Cryptography is derived from Greek words "kryptos" (hidden) and "graphein" (writing). It is the science and art of secure communication that protects information from unauthorized access.

Key Definitions:

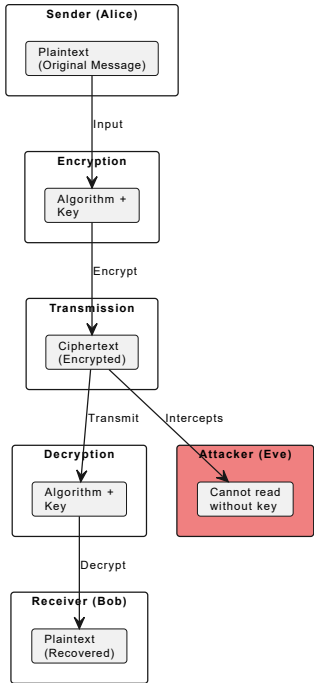
Cryptography: The study and practice of techniques for secure communication in the presence of adversaries. It involves creating and analyzing protocols that prevent third parties from reading private messages.

Encryption: The process of converting readable plaintext into unreadable ciphertext using an algorithm and a key. Only authorized parties with the correct key can decrypt and read the original message.

Decryption: The reverse process of encryption that converts ciphertext back to its original plaintext form using the appropriate key and algorithm.

Key: A piece of information (parameter) that determines the functional output of a cryptographic algorithm. The same algorithm with different keys produces different results.

Cryptographic Process Diagram:



Importance of Cryptography for Network Security:

Cryptography provides four fundamental security services (CIA + Non-repudiation):

1. **Confidentiality (Privacy):**

- Ensures that only authorized parties can access the information
- Prevents eavesdroppers from reading sensitive data
- Example: Encrypted emails, HTTPS websites, VPN tunnels

2. **Authentication:**

- Verifies the identity of communicating parties
- Confirms that the message originated from the claimed sender
- Example: Digital certificates, login credentials, digital signatures

3. **Integrity:**

- Ensures that data has not been altered during transmission
- Detects any unauthorized modifications to the message
- Example: Hash functions (MD5, SHA), Message Authentication Codes (MAC)

4. **Non-repudiation:**

- Prevents sender from denying they sent the message
- Provides proof of origin and delivery
- Example: Digital signatures, audit logs, timestamps

Types of Cryptographic Systems:

1. Symmetric Key Cryptography:

Concept: Same key is used for both encryption and decryption. Both sender and receiver must securely share this secret key.

Characteristics:

- Fast encryption/decryption
- Shorter key lengths (128-256 bits)
- Key distribution is challenging
- N users need N(N-1)/2 keys for pairwise communication

Common Algorithms:

Algorithm	Key Size	Block Size	Status
DES	56 bits	64 bits	Obsolete
3DES	168 bits	64 bits	Legacy
AES	128/192/256 bits	128 bits	Current standard
Blowfish	32-448 bits	64 bits	Secure
ChaCha20	256 bits	Stream	Modern, fast

2. Asymmetric Key Cryptography:

Concept: Uses a pair of mathematically related keys - a public key (shared openly) and a private key (kept secret). What one key encrypts, only the other can decrypt.

Characteristics:

- Solves key distribution problem
- Slower than symmetric encryption
- Longer key lengths (2048-4096 bits)
- Public key can be freely distributed
- Private key must be kept secure

Common Algorithms:

Algorithm	Based On	Key Size	Usage
RSA	Integer factorization	2048+ bits	Encryption, signatures
DSA	Discrete logarithm	1024+ bits	Digital signatures
ECC	Elliptic curves	256+ bits	Modern, efficient
Diffie-Hellman	Discrete logarithm	2048+ bits	Key exchange

Symmetric vs Asymmetric Comparison:

Feature	Symmetric	Asymmetric
Keys Used	One shared secret key	Public + Private key pair
Speed	Fast	Slow (100-1000x slower)
Key Length	Shorter (128-256 bits)	Longer (2048-4096 bits)
Key Distribution	Difficult (secure channel needed)	Easy (public key is public)
Scalability	Poor (N² keys for N users)	Good (2N keys for N users)
Usage	Bulk data encryption	Key exchange, digital signatures
Examples	AES, DES, 3DES	RSA, ECC, DSA
Security Basis	Key secrecy	Mathematical problems

Block Cipher vs Stream Cipher:

Feature	Block Cipher	Stream Cipher
Processing	Fixed-size blocks	Bit-by-bit or byte-by-byte
Speed	Slower	Faster
Padding	Required for incomplete blocks	Not needed
Error Propagation	Affects entire block	Affects only corrupted bit
Examples	AES, DES	RC4, ChaCha20, Salsa20
Modes	ECB, CBC, CTR, GCM	Continuous stream
Use Cases	File encryption, secure storage	Real-time communication

Hash Functions:

Purpose: Create fixed-size digital fingerprint of data for integrity verification.

Properties:

- One-way (cannot reverse)
- Deterministic (same input = same output)
- Collision-resistant (hard to find two inputs with same hash)
- Avalanche effect (small change = completely different hash)

Common Hash Algorithms:

Algorithm	Output Size	Status
MD5	128 bits	Broken, avoid
SHA-1	160 bits	Deprecated
SHA-256	256 bits	Secure, recommended
SHA-3	224-512 bits	Current standard

Cryptographic Applications in Networks:

Application	Cryptographic Techniques Used
HTTPS/SSL/TLS	RSA, AES, SHA, certificates
VPN	IPSec, AES, 3DES, IKE
Email (S/MIME)	RSA, AES, digital signatures
Wi-Fi (WPA2/3)	AES-CCMP, SAE
SSH	RSA, DSA, AES, SHA
Blockchain	SHA-256, ECDSA
Digital Signatures	RSA, DSA, ECDSA

Q2.1 Routing Algorithms

Question: Find shortest path using Dijkstra's algorithm OR Bellman-Ford algorithm from node A to all other nodes.

Introduction to Routing Algorithms:

Routing algorithms are fundamental to network layer operations, determining the optimal path for data packets to travel from source to destination. These algorithms build and maintain routing tables that guide packet forwarding decisions at each router.

Classification of Routing Algorithms:

Type	Algorithm	Characteristics
Distance Vector	Bellman-Ford, RIP	Share routing tables with neighbors
Link State	Dijkstra, OSPF	Share topology with all routers
Path Vector	BGP	Share path information

Dijkstra's Algorithm (Shortest Path First):

Concept: Dijkstra's algorithm finds the shortest path from a single source node to all other nodes in a weighted graph with non-negative edge weights. It uses a greedy approach, always selecting the unvisited node with the smallest known distance.

Algorithm Steps:

DIJKSTRA(Graph G, Source s): 1. INITIALIZATION: - Set Distance[s] = 0 - Set Distance[all other nodes] = ∞ - Set Previous[all nodes] = NULL - Create set Unvisited = {all nodes}

2. MAIN LOOP (repeat until Unvisited is empty): a. Select node u from Unvisited with minimum Distance[u] b. Remove u from Unvisited (mark as visited) c. For each neighbor v of u:

- Calculate: alt = Distance[u] + weight(u,v)
- If alt < Distance[v]:
 - Distance[v] = alt
 - Previous[v] = u

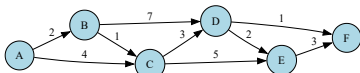
3. OUTPUT: Distance[] contains shortest distances Previous[] can reconstruct paths

Time Complexity:

- Using simple array: O(V²)
- Using binary heap: O((V + E) log V)
- Using Fibonacci heap: O(E + V log V)

Worked Example - Dijkstra's Algorithm:

Given Network Graph:



Step-by-Step Solution:

Step	Current Node	Visited Set	Distance[A]	Distance[B]	Distance[C]	Distance[D]	Distance[E]	Distance[F]
Init	-	{}	0	∞	∞	∞	∞	∞
1	A	{A}	0	2	4	∞	∞	∞
2	B	{A,B}	0	2	3	9	∞	∞
3	C	{A,B,C}	0	2	3	6	8	∞
4	D	{A,B,C,D}	0	2	3	6	8	7
5	F	{A,B,C,D,F}	0	2	3	6	8	7
6	E	{A,B,C,D,E}	0	2	3	6	8	7

Detailed Iteration Explanation:

Step 1 (Process A):

- Select A (distance = 0, minimum)
- Update neighbors: B = 0+2=2, C = 0+4=4
- Mark A as visited

Step 2 (Process B):

- Select B (distance = 2, minimum among unvisited)
- Update neighbors: C = min(4, 2+1) = 3, D = 2+7 = 9
- Mark B as visited

Step 3 (Process C):

- Select C (distance = 3, minimum among unvisited)
- Update neighbors: D = min(9, 3+3) = 6, E = 3+5 = 8
- Mark C as visited

Step 4 (Process D):

- Select D (distance = 6, minimum among unvisited)
- Update neighbors: E = min(8, 6+2) = 8 (no change), F = 6+1 = 7
- Mark D as visited

Step 5 (Process F):

- Select F (distance = 7, minimum among unvisited)
- No unvisited neighbors to update
- Mark F as visited

Step 6 (Process E):

- Select E (distance = 8, minimum among unvisited)
- F already visited
- Mark E as visited

Final Shortest Paths from A:

Destination	Shortest Distance	Path
A	0	A
B	2	A → B
C	3	A → B → C
D	6	A → B → C → D
E	8	A → B → C → E
F	7	A → B → C → D → F

Bellman-Ford Algorithm:

Concept: The Bellman-Ford algorithm computes shortest paths from a single source to all other vertices. Unlike Dijkstra's algorithm, it can handle graphs with negative edge weights and can detect negative weight cycles.

Algorithm Steps:

BELLMAN-FORD(Graph G, Source s): 1. INITIALIZATION: - Set Distance[s] = 0 - Set Distance[all other nodes] = ∞ 2. RELAXATION (repeat |V| - 1 times): For each edge (u, v) with weight w: If Distance[u] + w < Distance[v]: Distance[v] = Distance[u] + w 3. NEGATIVE CYCLE DETECTION: For each edge (u, v) with weight w: If Distance[u] + w < Distance[v]: Report "Negative cycle exists" 4. OUTPUT: Distance[] array

Time Complexity: O(V × E)

Worked Example - Bellman-Ford Algorithm:

Given Network (Same as above):

Edges: (A,B,2), (A,C,4), (B,C,1), (B,D,7), (C,D,3), (C,E,5), (D,E,2), (D,F,1), (E,F,3)

Iteration Table:

Iteration	Distance[A]	Distance[B]	Distance[C]	Distance[D]	Distance[E]	Distance[F]
Init	0	∞	∞	∞	∞	∞
1	0	2	3	6	8	7
2	0	2	3	6	8	7
3	0	2	3	6	8	7
4	0	2	3	6	8	7
5	0	2	3	6	8	7

Note: Algorithm converges after first iteration for this graph (no negative edges)

Comparison: Dijkstra vs Bellman-Ford:

Feature	Dijkstra	Bellman-Ford
Negative Weights	Not supported	Supported
Negative Cycles	Cannot detect	Can detect
Time Complexity	O(V²) or O(E log V)	O(V × E)
Approach	Greedy	Dynamic Programming
Usage	OSPF protocol	RIP, BGP protocols
Distributed Version	Complex	Easy (Distance Vector)
Convergence	Faster	Slower

Q2.2 TCP Connection Management (CRITICAL - 20 Marks)

Question: Explain TCP three-way handshake. Show normal operation and delayed packet scenarios.

Introduction to TCP:

Transmission Control Protocol (TCP) is a connection-oriented, reliable transport layer protocol. Before any data transfer, TCP establishes a logical connection between sender and receiver through a process called the three-way handshake.

TCP Segment Header Format:

Field	Size	Description
Source Port	16 bits	Sender's port number
Destination Port	16 bits	Receiver's port number
Sequence Number	32 bits	Position of first data byte
Acknowledgment Number	32 bits	Next expected byte
Header Length	4 bits	Header size in 32-bit words
Flags	6 bits	SYN, ACK, FIN, RST, PSH, URG
Window Size	16 bits	Receiver's buffer capacity
Checksum	16 bits	Error detection
Urgent Pointer	16 bits	Points to urgent data

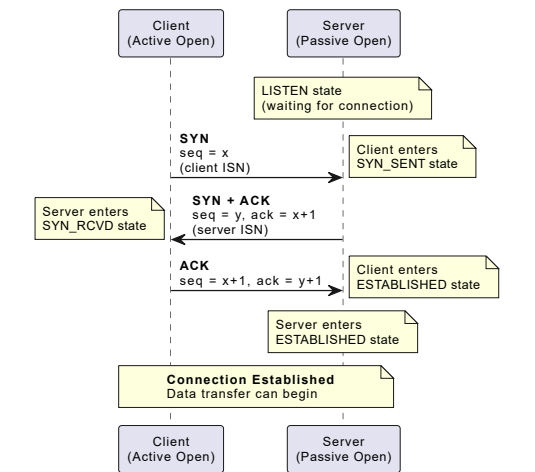
TCP Flags Explained:

Flag	Name	Purpose
SYN	Synchronize	Initiate connection, synchronize sequence numbers
ACK	Acknowledgment	Confirms receipt of data
FIN	Finish	Terminate connection gracefully
RST	Reset	Abort connection immediately
PSH	Push	Send data immediately to application
URG	Urgent	Contains urgent data

Three-Way Handshake (Connection Establishment):

Purpose: Establish a reliable connection, synchronize sequence numbers, and negotiate parameters.

Process Diagram:



Detailed Step Explanation:

Step 1: SYN (Synchronize)

- Client initiates connection by sending SYN segment
- Client chooses Initial Sequence Number (ISN) = x
- SYN flag = 1, ACK flag = 0
- Client state: CLOSED → SYN_SENT

Step 2: SYN + ACK (Synchronize + Acknowledge)

- Server acknowledges client's SYN with ACK
- Server sends its own SYN with its ISN = y
- SYN flag = 1, ACK flag = 1
- Acknowledgment number = x + 1 (next expected byte)
- Server state: LISTEN → SYN_RCVD

Step 3: ACK (Acknowledge)

- Client acknowledges server's SYN
- Sequence number = x + 1
- Acknowledgment number = y + 1
- SYN flag = 0, ACK flag = 1
- Both states: ESTABLISHED

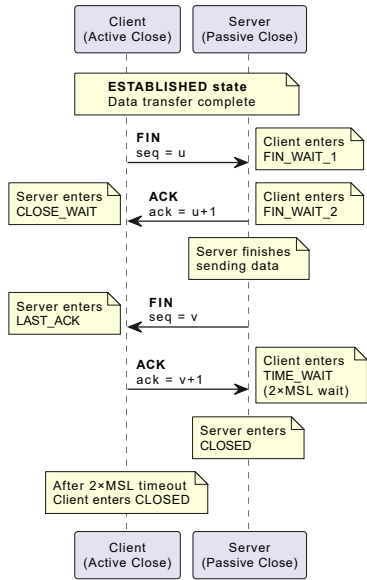
Why Three-Way? Why Not Two-Way?

- Prevents old duplicate connection requests from creating confusion
- Both sides confirm they can send and receive
- Synchronizes sequence numbers in both directions

Connection Termination (Four-Way Handshake):

TCP uses a four-way handshake for connection termination because the connection is full-duplex (each direction closed independently).

Process Diagram:



Termination Steps:

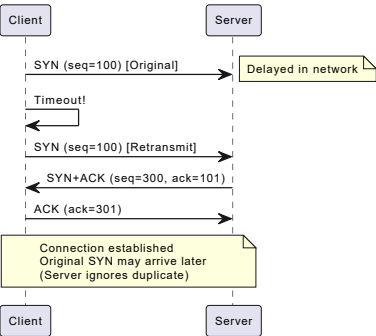
Step	Sender	Segment	Description
1	Client	FIN	Client wants to close, no more data to send
2	Server	ACK	Server acknowledges FIN
3	Server	FIN	Server finished, ready to close
4	Client	ACK	Client acknowledges, waits 2×MSL

TIME_WAIT State:

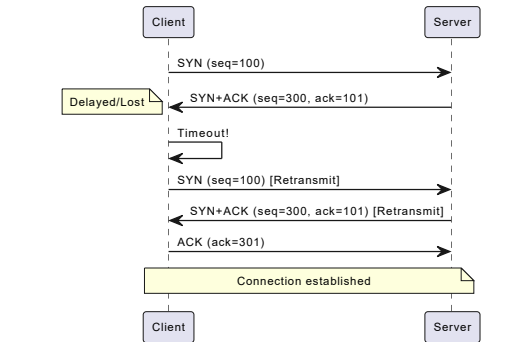
- Client waits for 2 × MSL (Maximum Segment Lifetime)
- Typically MSL = 2 minutes, so wait = 4 minutes
- Ensures final ACK reaches server
- Allows old duplicate segments to expire

Delayed Packet Scenarios:

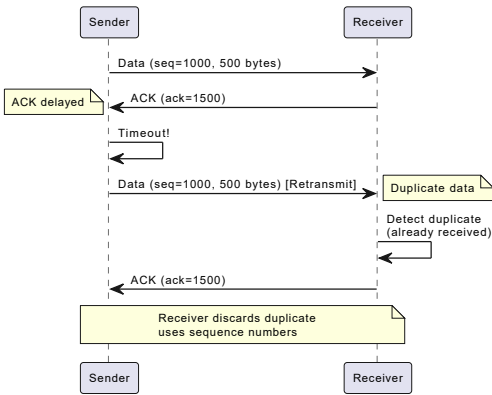
Scenario 1: SYN Packet Delayed



Scenario 2: SYN+ACK Delayed



Scenario 3: Data ACK Delayed



TCP State Diagram Summary:

State	Description
CLOSED	No connection exists
LISTEN	Server waiting for connection
SYN_SENT	Client sent SYN, awaiting SYN+ACK
SYN_RCVD	Server received SYN, sent SYN+ACK
ESTABLISHED	Connection active, data transfer
FIN_WAIT_1	Sent FIN, awaiting ACK
FIN_WAIT_2	Received ACK for FIN
CLOSE_WAIT	Received FIN, application notified
LAST_ACK	Sent FIN, awaiting final ACK
TIME_WAIT	Waiting 2×MSL before close
CLOSING	Both sides sent FIN simultaneously

Q2.3 Circuit vs Packet Switching (CRITICAL - 15-20 Marks)

Question: Compare circuit switching and packet switching. Explain virtual circuit vs datagram approach.

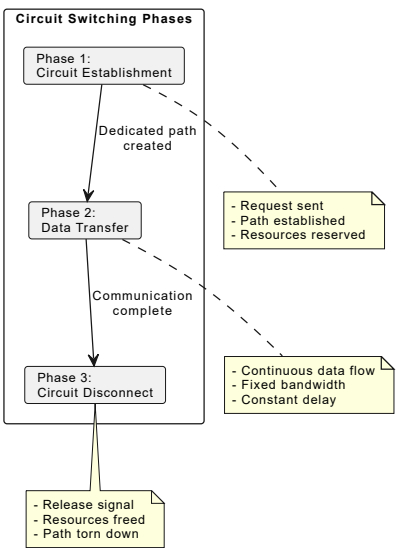
Introduction to Switching Techniques:

Switching is the mechanism by which data is transferred from source to destination through intermediate nodes in a network. The choice of switching technique affects network performance, reliability, and resource utilization.

Circuit Switching:

Definition: A dedicated communication path is established between sender and receiver for the entire duration of the communication session. Resources are reserved and remain allocated even when no data is being transmitted.

Three Phases of Circuit Switching:



Characteristics of Circuit Switching:

1. **Dedicated Path:** Exclusive channel between endpoints
2. **Resource Reservation:** Bandwidth reserved for entire session
3. **Fixed Delay:** Constant transmission delay
4. **Ordered Delivery:** Data arrives in order sent
5. **Inefficient:** Resources wasted during silent periods

Examples: Traditional telephone networks (PSTN), ISDN

Packet Switching:

Definition: Data is divided into packets that are transmitted independently through the network. Each packet may take a different route and is reassembled at the destination.

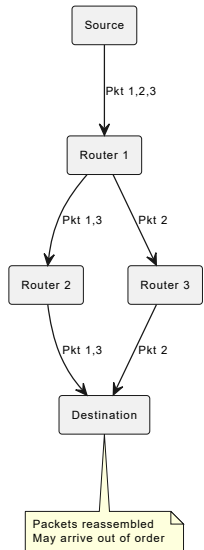
Characteristics of Packet Switching:

1. **Store-and-Forward:** Packets stored at each node before forwarding
2. **No Dedicated Path:** Packets may take different routes
3. **Statistical Multiplexing:** Bandwidth shared among users
4. **Variable Delay:** Each packet experiences different delay
5. **Efficient:** Better utilization of network resources

Types of Packet Switching:

1. **Datagram (Connectionless)**
2. **Virtual Circuit (Connection-oriented)**

Packet Switching Diagram:



Comprehensive Comparison Table:

Feature	Circuit Switching	Packet Switching
Path	Dedicated, fixed	Shared, dynamic
Connection Setup	Required before data transfer	Not required (datagram)
Resource Allocation	Reserved throughout	On-demand
Bandwidth Usage	Inefficient (idle time wasted)	Efficient (statistical mux)
Delay	Fixed, predictable	Variable, unpredictable
Congestion	Blocking during setup	May cause packet loss
Order of Arrival	Guaranteed in-order	May be out-of-order
Data Unit	Continuous bit stream	Discrete packets
Charging	Based on time/distance	Based on data volume
Transparency	Bit rate/coding transparent	Protocol specific
Reliability	High (dedicated path)	Depends on protocol
Overhead	Setup delay	Header overhead per packet
Examples	PSTN, ISDN	Internet, LAN
Best For	Voice, real-time	Data, bursty traffic

Virtual Circuit vs Datagram:

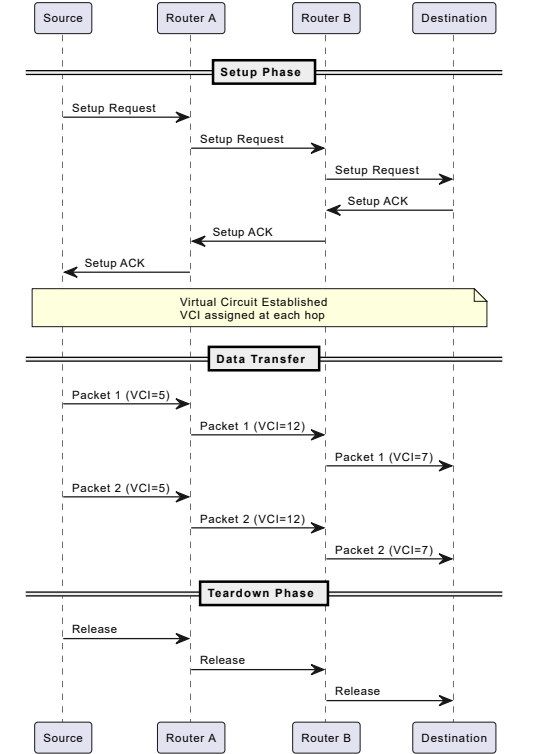
Virtual Circuit (Connection-Oriented Packet Switching):

Concept: A logical path is established before data transfer. All packets follow the same path and arrive in order. Combines features of circuit and packet switching.

Characteristics:

- Connection setup required (like circuit switching)
- Packets share links (like packet switching)
- Fixed path for all packets in a session
- Virtual Circuit Identifier (VCI) used for routing
- In-order delivery guaranteed

Virtual Circuit Process:



Datagram (Connectionless Packet Switching):

Concept: Each packet is treated independently with no prior connection setup. Packets may take different routes and arrive out of order.

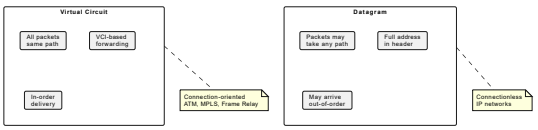
Characteristics:

- No connection setup
- Each packet routed independently
- Full destination address in every packet
- No guaranteed order
- More robust to failures

Virtual Circuit vs Datagram Comparison:

Feature	Virtual Circuit	Datagram
Connection Setup	Required	Not required
Address in Packet	Short VCI only	Full destination address
Routing Decision	Once at setup	Per packet
Path	Fixed for session	Dynamic per packet
Order Guarantee	In-order delivery	May arrive out-of-order
Failure Recovery	VC must be reestablished	Packets reroute automatically
State in Routers	Must maintain VC table	Stateless (except routing table)
Overhead	Setup/teardown overhead	Header overhead per packet
Delay	Lower (after setup)	Variable
Resource Reservation	Possible	Not possible
QoS Support	Easier	More difficult
Examples	ATM, MPLS, X.25	IP, Ethernet
Reliability	Higher	Lower (but flexible)

Diagram Comparison:



Q2.4 Error Detection Methods (HIGH PRIORITY - 15 Marks)

Question: Explain Hamming code. Calculate redundant bits for 7-bit character. Show CRC method.

Introduction to Error Detection and Correction:

In data communication, errors can occur due to noise, interference, or hardware failures. Error detection identifies if errors occurred, while error correction can fix some errors without retransmission.

Types of Errors:

Error Type	Description	Example
Single-bit Error	One bit changes	10110 → 10010
Burst Error	Multiple consecutive bits change	10110 → 11001

Error Control Methods:

Method	Capability	Example
Parity	Detect single-bit errors	Simple parity
Checksum	Detect multiple errors	Internet checksum
CRC	Detect burst errors	CRC-32
Hamming	Detect and correct single-bit	SECDED

Hamming Code - Detailed Explanation:

Concept: Hamming code is a linear error-correcting code that can detect up to two-bit errors and correct single-bit errors. It strategically places parity bits at positions that are powers of 2.

Formula for Redundant Bits:

$$2^r \geq m + r + 1$$

Where:

- **m** = number of data bits
- **r** = number of redundant (parity) bits

Calculating Redundant Bits for 7-bit Data:

r	2 ^r	m + r + 1	Condition Met?
1	2	7+1+1=9	2 ≥ 9? No
2	4	7+2+1=10	4 ≥ 10? No
3	8	7+3+1=11	8 ≥ 11? No
4	16	7+4+1=12	16 ≥ 12? Yes

Answer: For 7 data bits, r = 4 parity bits are required. **Total codeword length:** 7 + 4 = 11 bits

Parity Bit Positions:

Parity bits are placed at positions that are powers of 2: **1, 2, 4, 8, 16, ...**

Position Layout for 11-bit Codeword:

Position	1	2	3	4	5	6	7	8	9	10	11
Type	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7

Where: P = Parity bit, D = Data bit

Parity Bit Coverage:

Each parity bit checks specific positions based on binary representation:

Parity Bit	Position	Checks Positions	Binary Pattern
P1	1	1,3,5,7,9,11	Bit 0 is 1
P2	2	2,3,6,7,10,11	Bit 1 is 1
P4	4	4,5,6,7	Bit 2 is 1
P8	8	8,9,10,11	Bit 3 is 1

Worked Example: Encode 1001101 using Hamming Code

Data bits (D1-D7): 1 0 0 1 1 0 1

Step 1: Position the bits

Position	1	2	3	4	5	6	7	8	9	10	11
Type	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7
Value	?	?	1	?	0	0	1	?	1	0	1

Step 2: Calculate Parity Bits (Even Parity)

- P1** covers positions 1,3,5,7,9,11: ? + 1 + 0 + 1 + 1 + 1 = even
• Sum of data = 1+0+1+1+1 = 4 (even) → **P1 = 0**
- P2** covers positions 2,3,6,7,10,11: ? + 1 + 0 + 1 + 0 + 1 = even
• Sum of data = 1+0+1+0+1 = 3 (odd) → **P2 = 1**
- P4** covers positions 4,5,6,7: ? + 0 + 0 + 1 = even
• Sum of data = 0+0+1 = 1 (odd) → **P4 = 1**
- P8** covers positions 8,9,10,11: ? + 1 + 0 + 1 = even
• Sum of data = 1+0+1 = 2 (even) → **P8 = 0**

Step 3: Final Hamming Code

Position	1	2	3	4	5	6	7	8	9	10	11
Value	0	1	1	1	0	0	1	0	1	0	1

Transmitted Codeword: 01110010101

Error Detection and Correction:

If received word is 01110010001 (error at position 9):

Check	Positions	Values	Sum	Syndrome
C1	1,3,5,7,9,11	0,1,0,1,0,1	3	1
C2	2,3,6,7,10,11	1,1,0,1,0,1	4	0
C4	4,5,6,7	1,0,0,1	2	0
C8	8,9,10,11	0,0,0,1	1	1

Syndrome = C8 C4 C2 C1 = 1001 = 9 (decimal)

Error is at position 9! Flip bit at position 9 to correct.

CRC (Cyclic Redundancy Check) - Review:

CRC uses polynomial division for powerful burst error detection. (See Q1.1 for detailed CRC calculation)

CRC vs Hamming Comparison:

Feature	Hamming Code	CRC
Type	Error correction	Error detection
Capability	Correct 1-bit, detect 2-bit	Detect burst errors
Overhead	More redundant bits	Efficient
Complexity	Moderate	Simple division
Usage	Memory systems, RAID	Network protocols
Standard	SECDED	CRC-16, CRC-32

Q2.5 CSMA Protocols (HIGH PRIORITY - 15 Marks)

Question: Explain CSMA/CD, back-off algorithm. Discuss hidden station and exposed station problems.

Introduction to CSMA:

Carrier Sense Multiple Access (CSMA) is a media access control protocol where a station listens to the channel before transmitting. This reduces collisions but cannot eliminate them entirely due to propagation delay.

Types of CSMA:

Type	Action when Channel Busy	Action when Channel Idle
1-Persistent	Wait and continuously sense	Transmit immediately (p=1)
Non-persistent	Wait random time, then sense	Transmit immediately
P-persistent	Wait for next slot	Transmit with probability p

CSMA/CD (Collision Detection):

Concept: CSMA/CD adds collision detection to basic CSMA. Stations monitor the channel while transmitting, and if a collision is detected, they stop immediately and invoke a back-off algorithm.

Algorithm Steps:

CSMA/CD Algorithm: 1. Carrier Sense: Listen to channel 2. If channel BUSY: - Wait until idle - Wait IFG (Inter-Frame Gap)

1. If channel IDLE:
2. Begin transmission
3. Monitor channel while transmitting
4. If COLLISION detected:
5. Stop transmission
6. Send JAM signal (32 bits)
7. Increment collision counter
8. Execute back-off algorithm
9. Retry from step 1
10. If NO collision:
11. Transmission successful
12. Reset collision counter

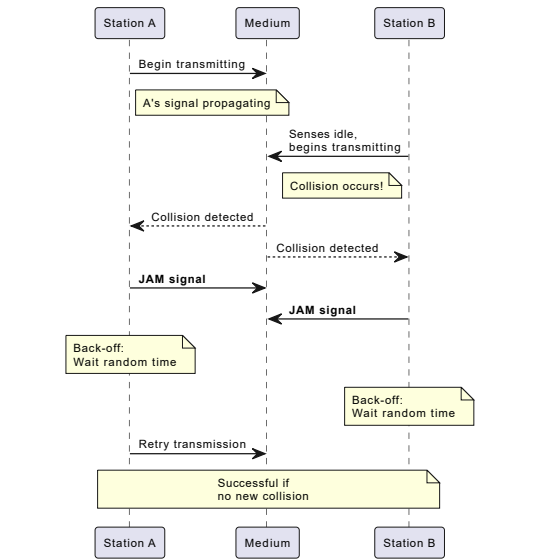
Minimum Frame Size Requirement:

For collision detection to work, frame transmission must take longer than the round-trip propagation time:

$$T_{frame} \geq 2 \times T_{propagation}$$
$$\frac{Frame_Size}{Bandwidth} \geq \frac{2 \times Distance}{Signal_Speed}$$

For Ethernet: Minimum frame = 64 bytes

CSMA/CD Timing Diagram:



Binary Exponential Back-off Algorithm:

Purpose: After a collision, this algorithm determines how long a station should wait before retransmitting. The waiting time increases exponentially with each successive collision.

Algorithm:

- Binary Exponential Back-off:
1. After nth collision (n starts at 1):
 - Define $k = \min(n, 10)$
 - Choose random number r from {0, 1, 2, ..., 2^k - 1}
 - Wait time = $r \times \text{slot_time}$
 2. Slot time = 51.2 μs (for 10 Mbps Ethernet)
 3. If n > 16:
 - Report failure to upper layer
 - Abort transmission

Example Calculations:

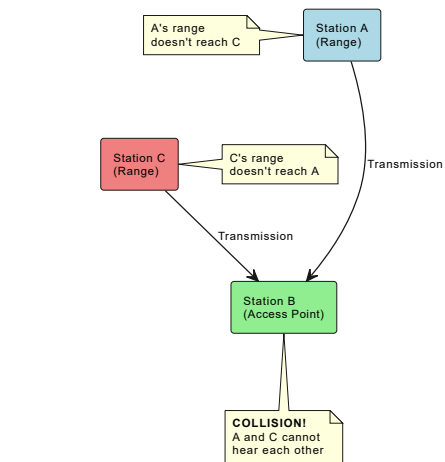
Collision #	k	Range (0 to 2 ^k - 1)	Possible Wait (slots)
1	1	{0, 1}	0 or 1
2	2	{0, 1, 2, 3}	0 to 3
3	3	{0, 1, ..., 7}	0 to 7
5	5	{0, 1, ..., 31}	0 to 31
10	10	{0, 1, ..., 1023}	0 to 1023
15	10	{0, 1, ..., 1023}	0 to 1023 (capped)

Properties:

- First collision: Short wait (fairness for light load)
- Many collisions: Longer wait (reduces congestion)
- Cap at k=10 prevents excessive waiting
- Failure after 16 attempts

Hidden Station Problem:

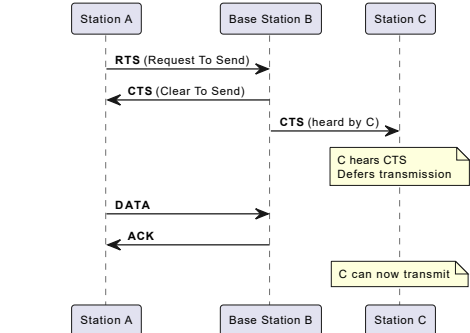
Scenario: Two stations that cannot hear each other both transmit to a common receiver, causing collision at the receiver.



Problem Explanation:

1. Station A transmits to Base Station B
2. Station C cannot sense A's transmission (out of range)
3. C senses channel idle and transmits to B
4. Collision occurs at B
5. Both transmissions are lost

Solution: RTS/CTS Mechanism

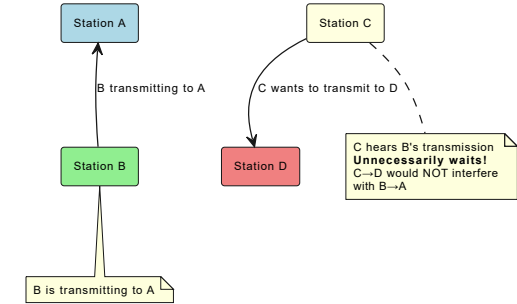


RTS/CTS Mechanism:

- Sender sends RTS (Request To Send) with duration
- Receiver replies with CTS (Clear To Send) with duration
- All stations hearing CTS defer transmission
- Solves hidden station problem

Exposed Station Problem:

Scenario: A station refrains from transmitting even though its transmission would not cause interference.



Problem Explanation:

1. Station B is transmitting to Station A
2. Station C wants to transmit to Station D
3. C hears B's transmission and senses channel busy
4. C waits unnecessarily
5. Actually, C→D would not collide with B→A

Why It's a Problem:

- Reduces network throughput
- Unnecessary waiting
- Wastes transmission opportunities

Comparison:

Problem	Hidden Station	Exposed Station
Cause	Stations can't hear each other	Station hears irrelevant transmission
Effect	Collision at receiver	Unnecessary waiting
Result	Data loss	Throughput reduction
Solution	RTS/CTS	More complex MAC

CSMA/CA (Collision Avoidance):

Used in wireless networks (IEEE 802.11) because collision detection is difficult in wireless medium.

Differences from CSMA/CD:

Feature	CSMA/CD	CSMA/CA
Medium	Wired (Ethernet)	Wireless (Wi-Fi)
Collision Handling	Detect and abort	Avoid before transmission
Mechanism	Jam signal	RTS/CTS, ACK
IFS	IFG	DIFS, SIFS
Back-off	After collision	Before transmission

Q2.6 Wireless Networks (MEDIUM-HIGH PRIORITY - 15 Marks)

Question: Explain IEEE 802.11 protocol. Describe WSN topologies and applications.

IEEE 802.11 (Wi-Fi) Protocol:

Overview: IEEE 802.11 is a set of standards for implementing wireless local area network (WLAN) communication. It defines the MAC (Media Access Control) and PHY (Physical) layer specifications.

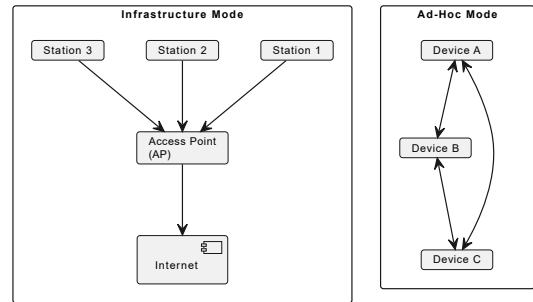
IEEE 802.11 Standards Evolution:

Standard	Frequency	Max Speed	Range	Year
802.11a	5 GHz	54 Mbps	35m	1999
802.11b	2.4 GHz	11 Mbps	38m	1999
802.11g	2.4 GHz	54 Mbps	38m	2003
802.11n	2.4/5 GHz	600 Mbps	70m	2009
802.11ac	5 GHz	6.9 Gbps	35m	2013
802.11ax	2.4/5/6 GHz	9.6 Gbps	35m	2019

802.11 Architecture:

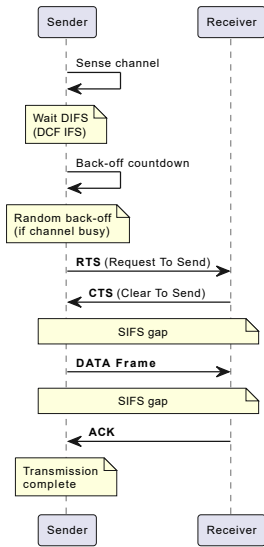
Two Modes of Operation:

1. **Infrastructure Mode (BSS - Basic Service Set)**
2. Stations communicate through Access Point (AP)
3. AP connected to wired network
4. Most common configuration
5. **Ad-Hoc Mode (IBSS - Independent BSS)**
6. Stations communicate directly (peer-to-peer)
7. No access point required
8. Temporary networks



802.11 MAC Layer - CSMA/CA:

DCF (Distributed Coordination Function):



Inter-Frame Spacing (IFS):

IFS Type	Duration	Purpose
SIFS (Short)	Shortest	ACK, CTS, data fragments
PIFS (PCF)	Medium	Point Coordination Function
DIFS (DCF)	Longest	New transmissions

802.11 Frame Format:

Field	Size	Description
Frame Control	2 bytes	Type, subtype, flags
Duration/ID	2 bytes	NAV (Network Allocation Vector)
Address 1	6 bytes	Receiver address
Address 2	6 bytes	Transmitter address
Address 3	6 bytes	BSSID or destination
Sequence Control	2 bytes	Fragment/sequence numbers
Address 4	6 bytes	Used in WDS
Data	0-2312 bytes	Payload
FCS	4 bytes	Frame Check Sequence

Wireless Sensor Networks (WSN):

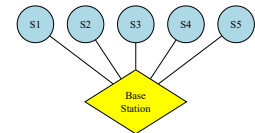
Definition: A WSN consists of spatially distributed autonomous sensors that monitor physical or environmental conditions and cooperatively pass data through the network to a main location (sink node).

WSN Components:

Component	Function
Sensor Node	Sensing, processing, communication
Sink/Base Station	Collects data from all sensors
Gateway	Connects WSN to external networks
User Interface	Visualization and control

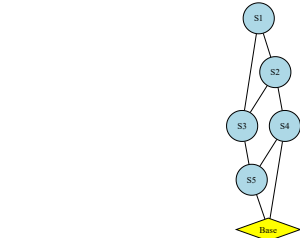
WSN Topologies:

1. Star Topology:



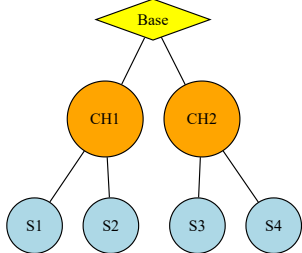
- All sensors directly communicate with base station
- Simple, low latency
- Limited range, single point of failure

2. Mesh Topology:



- Multi-hop communication
- Self-healing, robust
- Higher complexity and energy consumption

3. Tree/Cluster Topology:



- Cluster heads aggregate data
- Energy efficient (reduces transmissions)
- Hierarchical organization

WSN Topology Comparison:

Feature	Star	Mesh	Tree/Cluster
Range	Limited	Extended	Extended
Reliability	Low	High	Medium
Energy	Low (for sensors)	High	Medium
Latency	Low	Variable	Medium
Scalability	Limited	Good	Very good
Complexity	Simple	Complex	Moderate

WSN Applications:

Domain	Applications
Environmental	Forest fire detection, flood monitoring, air quality
Healthcare	Patient monitoring, elderly care, drug administration
Military	Battlefield surveillance, enemy tracking, intrusion detection
Industrial	Machine health monitoring, process control, inventory tracking
Smart City	Traffic monitoring, smart parking, waste management
Agriculture	Soil moisture, precision farming, livestock tracking
Home Automation	Security, energy management, appliance control

WSN Challenges:

Challenge	Description
Energy	Limited battery, no recharging
Bandwidth	Low data rate communication
Memory	Limited storage capacity
Security	Vulnerable to attacks
Scalability	Large number of nodes
Reliability	Node failures common

Q2.7 Multiplexing Techniques (MEDIUM PRIORITY - 10 Marks)

Question: What is multiplexing? Explain FDM, TDM, and upward multiplexing.

Introduction to Multiplexing:

Definition: Multiplexing is a technique that combines multiple signals for transmission over a single communication channel. It maximizes the use of expensive high-bandwidth channels by sharing them among multiple users.

Why Multiplexing?

- Efficient use of communication resources
- Cost reduction (share expensive infrastructure)

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- Unreliable (no acknowledgments)
- No flow control
- No congestion control
- Message-oriented (preserves message boundaries)
- Low overhead (8-byte header)

UDP Header Format:			
0	7	8	15
16	23	24	31
Source Port		Dest Port	
Length		Checksum	
Data			

Field	Size	Description
Source Port	16 bits	Sender's port number
Destination Port	16 bits	Receiver's port number
Length	16 bits	Total UDP datagram length (header + data)
Checksum	16 bits	Error detection (optional in IPv4)

Total Header Size: 8 bytes

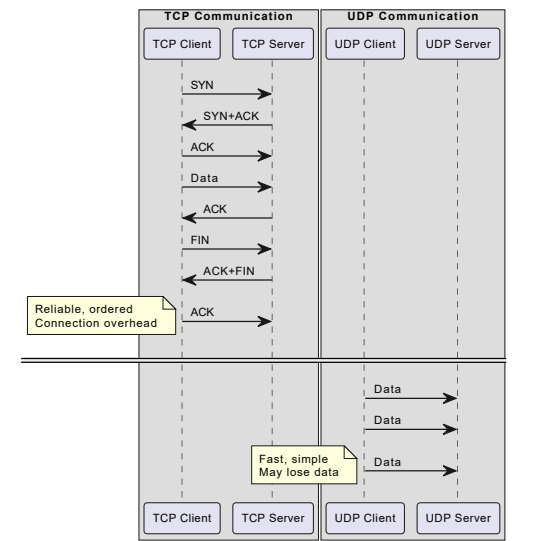
UDP Functionality:

1. Multiplexing/Demultiplexing:
- Uses port numbers to direct data to correct application
 - Source port: Identifies sending process
 - Destination port: Identifies receiving process
2. Error Detection:
- Optional checksum covers header and data
 - Uses pseudo-header (includes IP addresses)
 - Checksum = 0 means not computed
3. Message Boundaries:
- Preserves application message boundaries
 - Each send() results in one datagram
 - Each receive() gets one complete datagram

When to Use UDP:	
Application	Why UDP?
DNS	Small queries, quick response
DHCP	Broadcast discovery, simple transaction
SNMP	Network management, periodic updates
TFTP	Simple file transfer, implements own reliability
VoIP	Real-time, delay-sensitive
Video Streaming	Delay-sensitive, can tolerate loss
Online Gaming	Low latency required
NTP	Time synchronization

TCP vs UDP Comprehensive Comparison:		
Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery (ACKs, retransmission)	No guarantee (best effort)
Ordering	In-order delivery	No ordering guarantee
Flow Control	Yes (sliding window)	No
Congestion Control	Yes (slow start, AIMD)	No
Error Recovery	Automatic retransmission	None (application handles)
Header Size	20-60 bytes	8 bytes
Overhead	Higher	Lower
Speed	Slower (due to reliability)	Faster
Data Boundaries	Stream-oriented	Message-oriented
Broadcast/Multicast	No	Yes
State	Stateful (maintains connection)	Stateless
Full Duplex	Yes	Yes
Examples	HTTP, FTP, SMTP, SSH	DNS, DHCP, VoIP, Gaming

Diagram: TCP vs UDP Communication:



Q2.10 Network Security (MEDIUM PRIORITY - 10 Marks)

Question: Explain recent cyber attacks. Describe digital certificates. Explain DES algorithm.

Recent Cyber Attacks:

1. Ransomware Attacks:
- Definition:** Malware that encrypts victim's files and demands payment for decryption key.
- Notable Examples:**
- **WannaCry (2017):** Affected 200,000+ computers in 150 countries
 - **NotPetya (2017):** Caused \$10 billion in damages worldwide
 - **Colonial Pipeline (2021):** Disrupted fuel supply in US East Coast
 - **LockBit, REvil:** Major ransomware-as-a-service operations

- Attack Process:**
1. Initial infection (phishing, exploit)
 2. Encryption of files
 3. Ransom demand displayed
 4. Decryption key provided (if paid)

2. DDoS (Distributed Denial of Service):

Definition: Attack that overwhelms target servers with traffic from multiple sources, making services unavailable.

- Types:**
- **Volumetric:** Flood bandwidth (UDP flood, ICMP flood)
 - **Protocol:** Exploit protocol weaknesses (SYN flood)
 - **Application:** Target application layer (HTTP flood)

- Notable Attacks:**
- GitHub (2018): 1.35 Tbps attack
 - AWS (2020): 2.3 Tbps attack

3. Phishing:

Definition: Social engineering attack that tricks users into revealing sensitive information through fake emails, websites, or messages.

- Types:**
- **Spear Phishing:** Targeted at specific individuals
 - **Whaling:** Targeting executives
 - **Smishing:** Via SMS
 - **Vishing:** Via voice calls

4. Man-in-the-Middle (MitM):

Definition: Attacker secretly intercepts and possibly alters communication between two parties who believe they're communicating directly.

- Techniques:**
- ARP spoofing
 - DNS spoofing
 - SSL stripping

- Wi-Fi eavesdropping

Attack Summary Table:

Attack	Target	Method	Prevention
Ransomware	Files/Data	Encryption	Backups, updates
DDoS	Availability	Traffic flood	CDN, filtering
Phishing	Credentials	Social engineering	Training, 2FA
MitM	Communication	Interception	HTTPS, VPN

Digital Certificates:

Definition: A digital certificate is an electronic document that binds a public key to an identity. It is issued by a trusted Certificate Authority (CA).

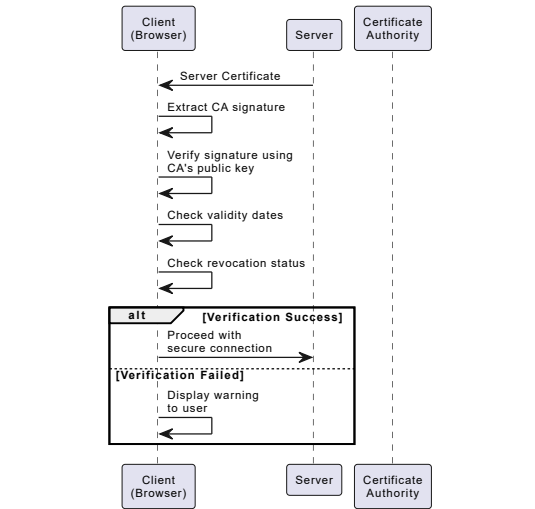
Purpose:

1. **Authentication:** Verify identity of entity
2. **Public Key Distribution:** Safely distribute public keys
3. **Trust Establishment:** Build chain of trust
4. **Secure Communication:** Enable HTTPS, secure email

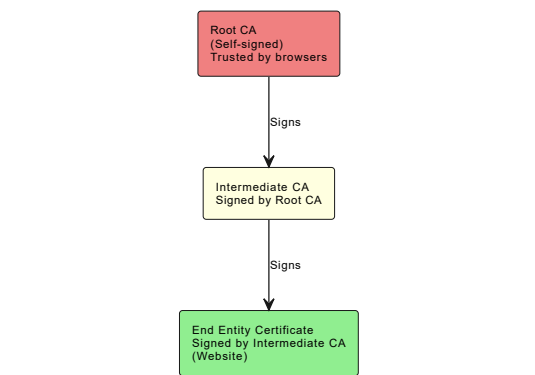
Digital Certificate Contents (X.509 Standard):

Field	Description
Version	Certificate format version (v1, v2, v3)
Serial Number	Unique identifier from CA
Signature Algorithm	Algorithm used to sign certificate
Issuer	CA that issued certificate
Validity Period	Not Before and Not After dates
Subject	Entity the certificate identifies
Subject Public Key	Public key being certified
Extensions	Additional info (key usage, etc.)
CA Signature	CA's digital signature

Certificate Verification Process:



Certificate Chain (Chain of Trust):



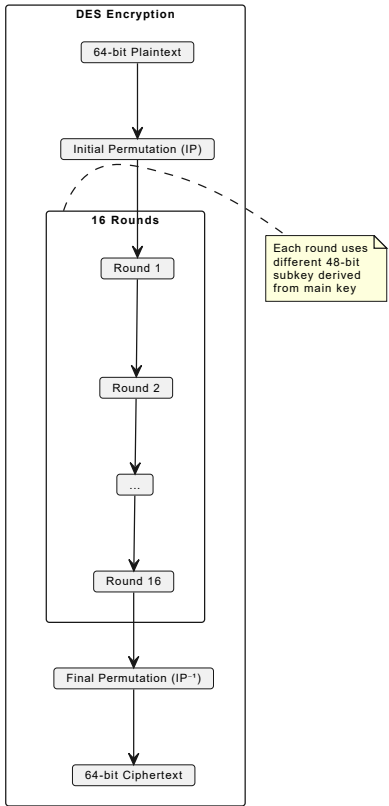
DES (Data Encryption Standard):

Overview: DES is a symmetric-key block cipher that was the federal standard for encryption from 1977 to 2002. It has been superseded by AES but remains important historically.

DES Parameters:

Parameter	Value
Block Size	64 bits
Key Size	64 bits (56 effective + 8 parity)
Number of Rounds	16
Structure	Feistel network
Status	Obsolete (use AES)

DES Algorithm Structure:



Feature	DES	AES
Security	Broken	Secure
Status	Obsolete	Current standard

DES Round Function (Feistel Function):

- Each round performs:
1. **Expansion:** 32 bits → 48 bits
 2. **Key Mixing:** XOR with 48-bit subkey
 3. **Substitution:** 8 S-boxes (6 bits → 4 bits each)
 4. **Permutation:** Bit permutation (P-box)

- Feistel Structure:
- Split 64-bit block into Left (32) and Right (32)
 - $L(i) = R(i-1)$
 - $R(i) = L(i-1) \text{ XOR } F(R(i-1), K(i))$

DES Weaknesses:	
Issue	Description
Key Size	56 bits too short (brute force possible)
Known Attacks	Differential and linear cryptanalysis
Weak Keys	Some keys produce identical subkeys
Complementation	$E(K,P)$ = complement of $E(K',P')$

DES Variants:		
Variant	Description	Key Size
2-Key 3DES	$E(K1, D(K2, E(K1, P)))$	112 bits
3-Key 3DES	$E(K1, D(K2, E(K3, P)))$	168 bits

AES vs DES Comparison:		
Feature	DES	AES
Block Size	64 bits	128 bits
Key Size	56 bits	128/192/256 bits
Rounds	16	10/12/14
Structure	Feistel	Substitution-Permutation
Speed	Slower	Faster